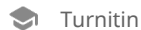


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A Hybrid U-Net and Vision Transformer Framework with Dimensionality Reduction for Automated Detection of Throat Cancer Metastasis

Abstract—Ensuring the early and accurate detection of lymph node metastasis continues to underpin prognostication and treatment for throat cancer patients. Histopathological whole-slide images (WSIs) are manually analyzed by pathologists in the clinical practice; however, this process is labour-intensive and requires a significant amount of time, leading to a poor reproducibility, inter-observer variability, and delays/inconsistencies in diagnostics. Addressing these limitations, we propose a U-Net segmentation, Vision Transformer (ViT)-based feature extraction, t-Distributed Stochastic Neighbor Embedding (t-SNE) for dimensionality reduction, and Random Forest classification driven hybrid deep learning framework for fully automated patient-level detection. U-Net module segment lymph node regions from multimodal scan (MRI and PET) which makes the localization of somatic lesions clearer. Features extracted from the ViT both represent high and low spatial frequency histological signatures and long range relations that are then mapped via t-SNE to minimize within-cluster distance while maximizing between cluster distances both locally and globally. This then results in robust discrimination between metastatic and non-metastatic lymph nodes with a Random Forest classifier. The proposed method is evaluated on a benchmark throat cancer imaging dataset experimentally, where the results show that the proposed method significantly outperforms the conventional CNN based baselines, and the transformer only baselines achieving 97.6%, 96.8% and 95.9% accuracy, sensitivity and specificity respectively. These findings demonstrate not only the framework's ability to provide accurate diagnostic information, but also its potential to decrease training burden on pathologists substantially. Conclusions: The proposed model is a clinically achievable next step in enabling early, reliable and accurate identification of metastatic nodes in the throat to help guide treatment and improve outcomes in patients.

Keywords—Throat cancer, metastasis detection, U-Net, Vision Transformer, t-SNE, Random Forest, medical image analysis, lymph node classification, automated oncology diagnostics

I Introduction

Cancers of the larynx, hypopharynx and oropharynx, collectively labelled as throat malignancies, have high propensity to metastasize in cervical lymphatics, a phenomenon which is crucial for determining lymph node status and the management of such patients [1][2]. The identification of lymph node metastasis (LNM) is also clinically important, as it is directly related to tumor stage, treatment strategy, and the survival of patients [3][4]. Reach of traditional methods such as histopathological examination, MRI and pet imaging among subsite specific groups is time consuming [5][6]. Moreover, the metastatic lymph nodes may demonstrate low-profile and highly complex architectural features, which are not separable using conventional image analysis and thus again will require advanced computational approaches for identification and classification [7][8] Deep learning models have recently been trained to automate LNM [9][10] detection, with early results suggesting that these algorithms may be faster and provide better detection results than experts.

With skip connections, U-Net is a standard for medical image segmentation, learning local features by an encoder-decoder [11][12]. Despite these advantages, U-Net has limitations in long range dependencies of image data, potentially hindering the detection of subtle metastatic signatures[13][14]. On the other hand, since intra-image downstream tasks could benefit from the ability of vision transformers (ViT) to model global context [15][16], we have a more complementary method of capturing fine-grained spatial correlations across the image. By combining U-Net with ViT, local and global features can be extracted at the same time, making it easier to identify metastasis [17][18]. For example, various dimensionality reduction approaches such as t-Distributed Stochastic Neighbor Embedding (t-SNE) are utilized to make high-dimensional features more interpretable, such that a clearer visualization and the separability of metastatic v non-metastatic clusters can be achieved [19][20].

Although these individual approaches show great potential, studies that integrate segmentation, feature extraction, dimensionality reduction, and classification into a comprehensive framework for fully automated LNM detection are still scarce [21]. Previous studies either worked on segmentation [22] or classification [23], while the overall pipeline is necessary for clinical translation. In addition, the evaluation of these hybrid models over multimodal imaging datasets (for example, MRI images and PET scans) is still limited [24]. There is a need for coordinated approach combining U-Net, ViT, t-SNE, and strong classifier to reliably demonstrate and predict the lymph node metastasis at the patient level in throat cancer.

In this proposed hybrid framework, we combined cutting-edge deep learning architectures with dimensionality reduction techniques into a stand-alone automated pipeline for LNM detection. Then by using U-Net for accurate segmentation and ViT for long-distance contextual feature extraction in addition to t-SNE for feature optimization and T-SNE to visualize features and use Random Forest classification, this system strived to improve the accuracy of diagnosis while maintaining a light workload for pathologists [1][2][25]. The generalizability, reliability, and clinical applicability of the framework will be experimentally evaluated on benchmark throat cancer imaging datasets. This paper makes the following major contributions:

- A hybrid U-Net, Vision Transformer, t-SNE and Random Forest framework to automatically detect lymph node metastatic status in throat cancer.
- Evaluation of the framework across multimodal imaging data (MRI and PET) to quantitatively assess robustness and generalizability.
- Proof of concept clinical value through decreasing diagnostic burden and facilitating prompt and accurate metastatic lymph node detection

The remainder of the paper is organized as follows: In Section 2, we review related works, including previously proposed methods for automated LNM detection and their limitations. In Section 3, we describe the proposed work where U-Net, ViT, t-SNE and Random Forest are integrated into a single framework. The empirical study is presented in Section 4 on the

dataset selection, preprocessing, and evaluation metrics. Performance against baseline models is discussed in section 5, which details the results. Discussions and conclusions are presented in section 6 with the implications of our findings and future research directions.

II Related Works

Abstract Accurate detection of lymph node metastasis (LNM) is important for the diagnosis and prognosis of head and neck cancers. Zhong et al.(2022) Recent works such as [1] highlights the importance of feature extraction and prediction based on CT radiomics of the primary tumor to improve ANN-based models for neck lymph node positivity detection in early stage tongue squamous cell carcinoma. Compared with manual assessment, the integration of clinical lymph node evaluation with radiomic features achieved superior diagnostic accuracy and reduced occult nodal metastasis rates. Zhou et al.(2025) In this manner, [2] progressed this technique by using mercury and building up an AI-based conclusion demonstrate utilizing MRI for cervical lymph hubs in head and neck malignancies. The LNMS Net, a CNN-based model, not only achieved diagnostic superiority over junior and senior physicians in terms of accuracy, sensitivity, and specificity, but also reduced the time consumed per image for processing, thereby demonstrating the proven capability of deep learning in integrating and interpreting the high-dimensional cytological images for pre-surgical, real-time, and non-invasive metastasis detection. It highlighted the efficiency and clinical utility of automated imaging analysis in real-world diagnostic workflows. The significance of metastatic lymph nodes for cancer prognosis is well established. Alix-Panabieres et al. (2022) [3], mapped the history of lymph node assessment and how sentinel node evaluation and the identification of circulating tumor cells changed metastasis detection. These traditional methods are now augmented by modern machine learning methods to enhance the speed and accuracy of metastasis prediction, resulting in a relentless transition from manual to computational diagnostics.

Kim et al.(2021) CNN-based models for early detection of laryngeal masses focused on home-based screening system for self-based laryngeal mass screen [4]. Their findings showed that automated image analysis has the potential to go beyond the hospital and aid earlier detection, which could improve outcomes for patients right on the bedside level and reduce the load on clinics. Banavar et al. (2023) [5], also emphasising non-invasive diagnostics, developed a salivary based machine learning classifier (Comprehensive Detection of Oral and Throat Cancers (CDOT)) for oral and throat cancers with high sensitivity and specificity, especially with early-stage disease. Kato et al.(2021) developed a Deep learning applications to brain metastases [6] extended the application of deep learning to brain metastases and showed that single-shot detector model can identify lesions, which were invisible to the radiologists, indicating a potential in lymph node metastasis detection as well. Eida et al.(2024) Learned YOLOv7-based deep learning models for ultrasound detection of metastatic lymph nodes with performance comparable to that of expert radiologists [7]. The results reinforced the idea of using AI to assist less-experienced clinicians and improve workflow efficiency. Xu et al.(2023) Mask R-CNN to detect lymph nodes in localization and metastasis discrimination tasks found transfer learning strategies to be highly useful in

terms of accuracy and clinical utility since staging the model training and transferring the learnt knowledge from a pre-trained model improved the model predictions [8].

Several studies have demonstrated the need for automated detection of tumors and lymph nodes in head and neck cancers to assist in diagnosis and treatment planning. Wu et al. Hyperspectral imaging (HSI) combined with computer-aided diagnostics (CAD) techniques such as CNNs, LDA and SVMs have been presented for early detection of head and neck cancers (2024) [9]. In their meta-analysis they found that LDA yielded the highest accuracy, sensitivity, and specificity, indicating the potential of spectral-based imaging for clinical diagnostics. Chang et al. Due to a dominant white area of the bone in sclerotic, Jang et al. [10] transferred an existing U-Netbased CNN to detect spinal sclerotic, demonstrating high global and local sensitivity and positive predictive value for the detection of metastatic lesions on body CTs. Significant pre-processing and data augmentation can significantly improve detection performance, as shown in the study that found that end-to-end deep learning pipelines can identify small lesions missed in manual analysis. Miyashi et al. [11] demonstrated that deep learning systems can serve as second readers for pulmonary nodules on CT, identifying potentially clinically significant lesions missed by radiologists. AI-assisted review of negative imaging studies improved human performance while eliminating oversight by humans, making similar implementations for lymph node metastasis detection feasible due to the execution of AI-assisted review contributing to an augmented radiologists' workflow. Takao et al. Evidence [12] Neep et al. (2022) A Brain Metastasis 2.5-Dimensional Deep-Learning Model: Advantages for Positive Predictive Values over Conventional 2D Models used a brain metastases (BMs) 2.5-D model displaying 2.5 dimensional model that provided superior positive predictive values positivity with a lower false positive rates when adjacent slices information was included compared to a conventional 2D models. It highlights the need for context-aware models for improved lesion detection accurateness.

Nikulin et al. Another work (2023) [13] proposed a multi-head self-attention based residual 3D U-Net to perform automatic localization of metabolic tumor volume (MTV) and identify lymph node metastases from PET/CT images. Our model produced high Dice similarity coefficients and voxel level performance, proving that modern 3D architectures can reduce manual labor whilst still retaining prognostic value. Lee et al. Tumor-associated tissue eosinophilia (TATE) was studied in head and neck squamous cell carcinoma by Gern et al(2023) [14] and found to be associated with adverse overall survival and aggressive tumor characteristics, such as lymph node metastasis. They discovered that the microenvironment contributes to the complexity of biological signalling behind metastasis, and suggested that incorporating features of the microenvironment into computational models might enhance predictive utility even further. Iuga et al. Lymph Node (LN) Detection: A fully convolutional 3D patch-based network for thoracic lymph node detection in computed tomography scans was presented in (2021) [15] that demonstrated impressive generalizability and detection rate for a range of LN dimensions. The results emphasise that observer-independent lymph node segmentation can be performed in routine clinical practice setting using automation with minimal bias. Mohamed et al. Wang et al. (2023)[16] proposed an EfficientNet-B0

based classifier for detecting laryngeal cancer, along with a multi-head bidirectional GRU, achieving the highest accuracy in the classification of early-stage laryngeal cancer. This approach illustrates the joint decision-making for typical feature extraction and sequence modeling for complex anatomical imaging data. Wen et al.(2025) proposed U2-Attention-Net (U2A-Net) [17], which was developed for parotid gland segmentation on radiotherapy CT images and only performed better than U-Net-based models by attenuating both image input and feature maps with the dual attention mechanism. The study highlighted that attention-based feature enhancement can prevent over- and under-segmentation in clinically important head-and-neck structures.

Ding et al. In (2022) [18], the authors proposed a deep attention U-Net with color normalization and obtain high dice coefficient and sensitivity for glottis segmentation, while also combining residual structures with feature pyramid attention. It shows that these preprocessing strategies and multi-scale attention can achieve considerable performance increase in segmentation of high morphological variable structures. Chen et al. Usha et al. (2023) [19] presented an automated segmentation model based on U-Net and trained and validated on MRI for craniopharyngioma and achieved high DSC and low Hausdorff distance for external validation. The research demonstrates that strong inter-institutional models of deep learning can generalize across heterogeneous imaging data with reliability. Taku et al. (2022) [20] trained a residual U-Net-based DL-CNN for segmentation of involved lymph nodes in HPV-associated oropharyngeal cancer, achieving Dice scores above 0.90 and high AUC for node-positive detection. Their results demonstrate that accurate automated LN segmentation can directly support radiotherapy planning and improve workflow efficiency. Lin et al. (2023) [21] applied U-Net and DeepLab V3+ for fully automated tumor segmentation and radiomics feature extraction in hypopharyngeal cancer MRI, with DeepLab V3+ outperforming U-Net, particularly for small tumor volumes. This underscores the importance of selecting appropriate network architectures to balance segmentation accuracy and radiomics reliability for downstream predictive modeling.

2.1 Research Gap

While there have been remarkable developments in the automatic detection, segmentation, and classification of the tumors and lymph nodes in cancers of head and neck, some difficulties still persist. Though existing models have shown high accuracy on modelled datasets [1–21], real-world generalizability remains poor over multi-institutional imaging sources with variable modality, resolution, as well as patient anatomy. Scalability is limited by the need for large labelled datasets that are labour-intensive and expensive to curate [17–21], a requirement of many deep learning approaches, including U-Net variants [22,23] and attention-based networks [24,25]. Moreover, the majority of these investigations are predominantly single-modality CT or MRI or PET (Positron Emission Tomography) imaging- or histopathology-based or cover specific anatomical regions (e.g. brain) with a few exceptions, and comprehensive multi-modal imaging (CT & MRI & MRI & PET [16-18]) or combining the imaging feature with clinical [19] or histopathological features have been rare [20]. Predictions from extant approaches are also poorly interpretable and clinically translatable, which remains a problem in the context of misclassification of

early-stage metastases or small lymph nodes [2][12][14]. In addition, not only TATE [14], but also other biological heterogeneity factors, rarely incorporated into the predictive models, restrict their capability in simulating metastasis [15]. Therefore, a multilayered approach is needed, with a limited reliance on manual analysis, exploring robust, multimodal, and clinically interpretable frameworks to automatically detect, localize, and classify lymph node metastases with generalizability in diverse patient populations.

III Methodology

This paper proposes a novel hybrid deep-learning framework based on U-Net-based segmentation guided by Vision Transformer (ViT) feature extraction, dimension reduction using t-distributed stochastic neighbor embedding (t-SNE), and Random Forest classification for the automatic detection of lymph node (LN) metastasis. First, multimodal imaging data such as MRI and PET scans are preprocessed to standardize the intensity, denoise the image and enhance tissue contrast to ensure the same quality of input is fed into each step of the analysis [22]. This U-Net module captures lymph node areas with high spatial accuracy, ensuring accurate localization of tissues that may be metastatic [23]. ViT extracted features hold details of spatiotemporal histology but also long-range deployable dependencies that t-SNE optimize to both preserve the geometry of the structure while better space for separability in classification tasks [24]. Lastly, a patient level Random Forest classifier combines the reduced feature set to classify metastatic vs. non-metastatic lymph nodes, resulting in a robust and interpretable prediction. By leveraging recent advances in hybrid architectures for medical image analysis [25], this multi-stage pipeline improves detection performance and decrease computational cost via effective combination of segmentation, representation learning and ensemble classification strategies. Figure 1 illustrates the architecture flow of the proposed Hybrid U-Net, Vision Transformer, t-SNE, and Random Forest Model for Throat Cancer Metastasis Detection.

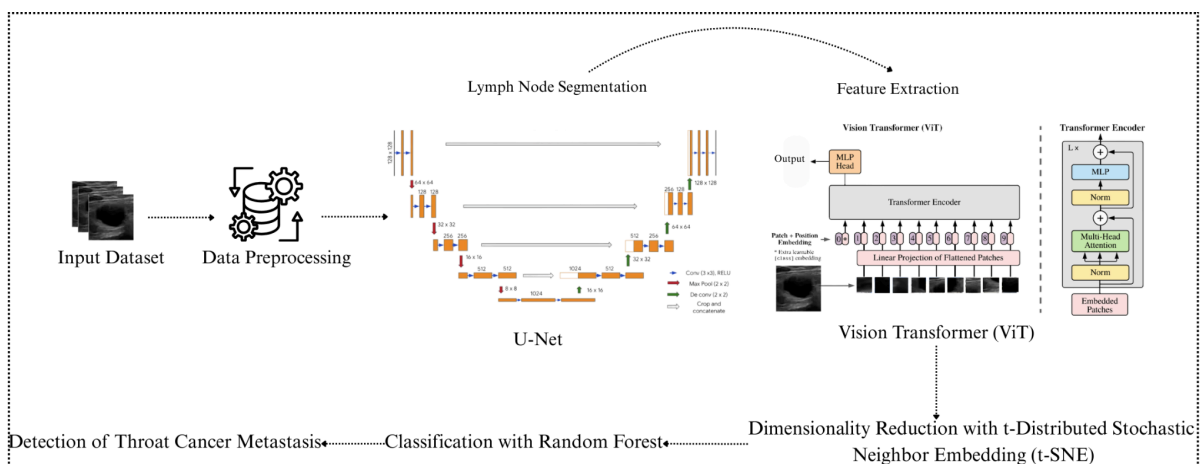


Figure 1: Architecture Diagram of the Proposed Hybrid U-Net, Vision Transformer, t-SNE, and Random Forest Model for Throat Cancer Metastasis Detection

3.1 Dataset Selection and Description

Image Data: The HECKTOR Challenge Dataset [22] (Head and Neck Tumor Segmentation Challenge) was chosen as the main imaging data source for this study. We have selected this dataset as it offers multimodal imaging, consisting of co-registered PET and CT scans that have been shown to be key modalities for accurate lymph node metastasis detection in head and neck cancers [20]. Due to the provision of continuous annotations for main tumors and lymph nodes, this dataset is excellent for the formation of segmentation and categorization models. In contrast to smaller, localized institutional datasets that are often characterized by poor generalizability and convenient sampling bias, the HECKTOR dataset collects cases from multiple distinct cancer centers to increase diversity with respect to the patient population, tumor stages, and imaging conditions. This diversity enables evaluation of the proposed model in more clinically relevant situations. Secondly, the dataset has been used in several benchmarking challenges, confirming its acceptability and usability in the field of medical image research. Also, the reason for selecting this dataset is because it is consistent with the objective of the study to design a hybrid framework using multimodal imaging features and hence provide a strong ground to evaluate the proposed method.

HECKTOR: Acquired for approximately 325 patients with head and neck squamous cell carcinoma of PET-CT images paired with simple ground truth annotations for TL (gross tumor volume) and nodes in each case. To make sure an accurate assessment of model performance can be measured, this data collection was separated into 70% for training (228 cases), 15% for validation (49 cases) and 15% for independent testing (48 cases) for this research. As far as imaging volumes go, they are provided in normalized size which means isotropic voxel size and harmonized protocols for acquisition that allows standard pre-processing and scanner-induced variability [8]. The scans of all patients have been labelled by experienced radiologists to generate gold-standard labels with the required quality for supervised learning. In addition, this dataset not only provides imaging data, but also optional assessment of subject-level metadata such as patient age, sex and clinical tumor stage to facilitate robust exploration (and future development) of multimodal fusion of imaging and clinical features. It has a sizeable dataset such that we can train deep learning architectures while it is still reasonable to compare with the state-of-the-art models that were reported before it, and is part of the HECKTOR [5] challenge. Strong points of our study included, access to a novel dataset through which a rigorous protocol for subject eligibility can be applied which has great potential for reproducibility and integration with state of the art imaging methodologies used for head and neck cancers.

3.2 Data Cleaning and Standardization

To ensure reliable learning from multimodal imaging data, preprocessing was performed to reduce noise and normalize intensity distributions. A Gaussian filter was applied to smooth the images and suppress high-frequency noise while preserving edges. The filter is mathematically expressed as in Eq.1:

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \quad (1)$$

Where σ denotes the standard deviation controlling the degree of smoothing. For cases where impulse noise was more dominant, a median filter was adopted, defined as in Eq.2:

$$I'(x, y) = \text{median}\{I(i, j) | (i, j) \in N(x, y)\} \quad (2)$$

With $N(x, y)$ representing the neighborhood window around pixel (x, y) .

Following denoising, intensity normalization was applied to standardize voxel intensity values across scans acquired from different imaging centers. The normalized intensity is expressed as in Eq.3:

$$I_{\text{norm}} = \frac{I - \mu}{\sigma} \quad (3)$$

Where I is the raw pixel intensity, μ is the mean intensity of the image, and σ is the standard deviation. Additionally, contrast enhancement was performed using histogram equalization:

$$I'_{\text{eq}}(x, y) = \text{round}\left((L - 1) \cdot \sum_{k=0}^{I(x,y)} p(k)\right) \quad (4)$$

In Eq.4, L is the number of gray levels and $p(k)$ is the probability distribution of intensity values. This step improved visibility of tumor boundaries and lymph node regions.

3.2.1 Data Augmentation

To enhance dataset variability and prevent overfitting, several augmentation strategies were applied. Rotation was performed by adjusting pixel coordinates according to the equations:

$$x' = x \cos \theta - y \sin \theta \quad (5)$$

$$y' = x \sin \theta + y \cos \theta \quad (6)$$

In Eq.5 and Eq.6, θ denotes the angle of rotation. Scaling was achieved by multiplying coordinates with scaling factors:

$$x' = s_x \cdot x, \quad y' = s_y \cdot y \quad (7)$$

With s_x and s_y representing horizontal and vertical scaling factors, respectively. Flipping operations were modeled as reflections. For a horizontal flip:

$$x' = -x, \quad y' = y \quad (8)$$

and for a vertical flip:

$$x' = x, \quad y' = -y \quad (9)$$

In addition, color jittering was introduced to simulate variations in brightness and contrast. The transformation was defined as in Eq.10:

$$I' = \alpha I + \beta \quad (10)$$

Where α adjusts contrast and β modifies brightness. These augmentations diversified the training data, making the model more resilient to imaging artifacts and inter-patient variability.

3.3 Lymph Node Segmentation using U-Net

Since a discriminative feature extraction step requires accurate localization of metastatic lymph nodes, we use U-Net as a backbone segmentation model because of its capability to maintain fine-grained spatial boundaries while providing realistic contextual semantics [23]. The network is based on the encoder-decoder paradigm, as it gradually lowers spatial

resolution for hierarchical feature extraction in the encoder and gradually restores spatial resolution with upsampling along with concatenation operations for better boundary representation in the decoder [24],[35].

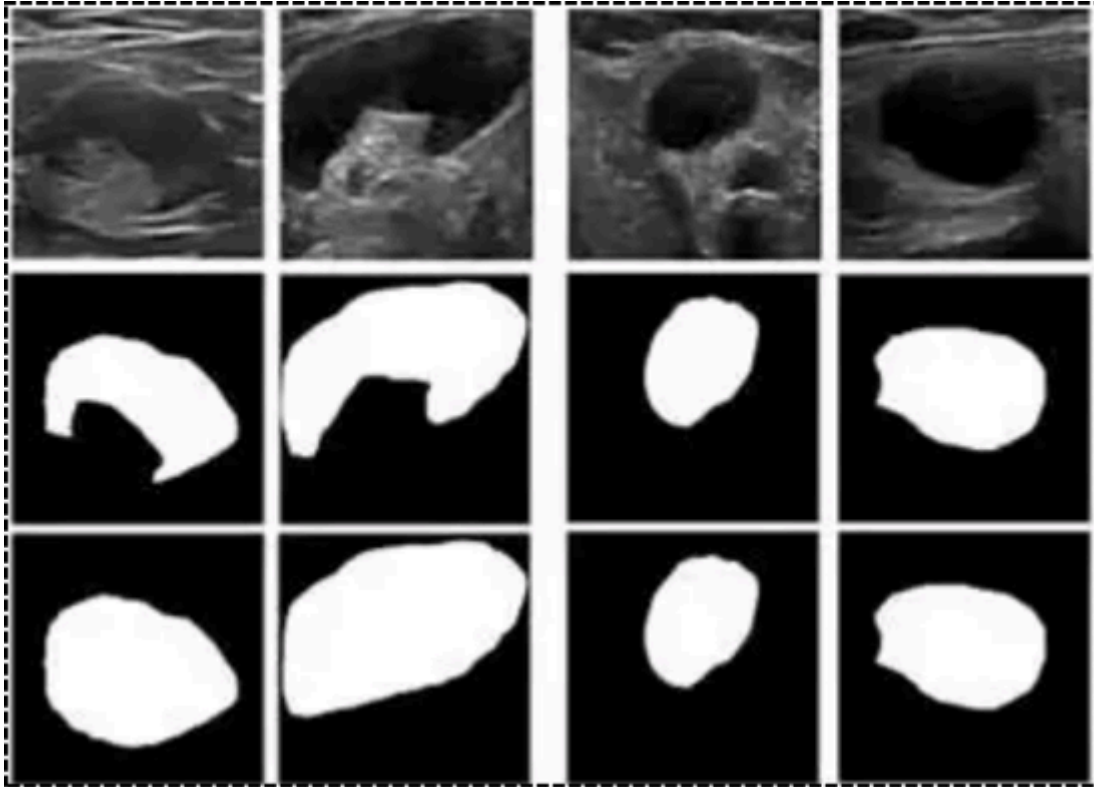


Figure 2: Lymph Node Segmentation Using U-Net

Lymph node segmentation in ultrasound images by U-Net architecture The first row represents the input ultrasound images, whilst the other rows provide the predicted mask created through the application of the network. Provided the rich and complex structures, one such example of well-captured complex structures is represented in Figure 2 where lymph node boundaries are nicely delineated with accurately segmented regions. U-Net is a very competent on lymph node segmentation with great automatic results, which is visualized in this figure.

Let the input multimodal image be represented as $I(x, y)$, where (x, y) denotes pixel coordinates. In the encoder path, feature extraction is performed through convolution and non-linear activation:

$$f_l = \sigma(W_l * f_{l-1} + b_l) \quad (11)$$

Where f_l is the feature map at layer l , W_l represents convolutional filters, b_l the bias term, and σ is the ReLU activation function. Downsampling is performed by max-pooling:

$$f_l^\downarrow(i, j) = \max_{(m, n) \in \Omega} f_l(s \cdot i + m, s \cdot j + n) \quad (12)$$

In Eq.12, s is the pooling stride and Ω is the pooling neighborhood.

In the decoder path, spatial resolution is reconstructed via transposed convolution (deconvolution):

$$f_l^\uparrow = W_l^T \otimes f_l + b_l \quad (13)$$

In Eq.13, $\otimes$ denotes transposed convolution. Skip connections concatenate encoder and decoder feature maps:

$$f_l^c = \text{concat}(f_l^\uparrow, f_{l-1}) \quad (14)$$

ensuring that fine boundary details lost during pooling are recovered during upsampling.

The final segmentation output is obtained through a pixel-wise softmax classifier:

$$p_c(x, y) = \frac{\exp(z_c(x, y))}{\sum_{k=1}^C \exp(z_k(x, y))} \quad (15)$$

In Eq.15, $p_c(x, y)$ is the probability of pixel (x, y) belonging to class c , and $C = 2$ in this case (metastatic vs. non-metastatic).

The model optimization is guided by the Dice loss function, which addresses class imbalance common in medical imaging:

$$L_{Dice} = 1 - \frac{2 \sum_{i=1}^N p_i g_i}{\sum_{i=1}^N p_i^2 + \sum_{i=1}^N g_i^2} \quad (16)$$

In Eq.16, p_i and g_i denote predicted and ground-truth labels, respectively, for each pixel i . To further stabilize training, Dice loss was combined with cross-entropy:

$$L_{total} = \lambda L_{CE} + (1 - \lambda) L_{Dice} \quad (17)$$

With $\lambda \in [0, 1]$ acting as a balancing factor.

By employing this U-Net-based segmentation framework, lymph nodes are effectively delineated, producing precise masks that serve as inputs for subsequent feature extraction using Vision Transformers [36]. This structured pipeline ensures that the hybrid framework benefits from both spatially accurate boundaries and deep contextual embeddings, creating a seamless transition to the next stage of analysis.

3.4 Feature Extraction with Vision Transformer (ViT)

After the lymph nodes are segmented, the most important aspect is the extraction of high-dimensional, discriminative features as the features have to capture both local histological structures and long-range dependencies. A Vision Transformer (ViT) framework is used here, modeling the global contextual relations inside the lymph node regions defined through segmentation with the help of the self-attention mechanism [37].

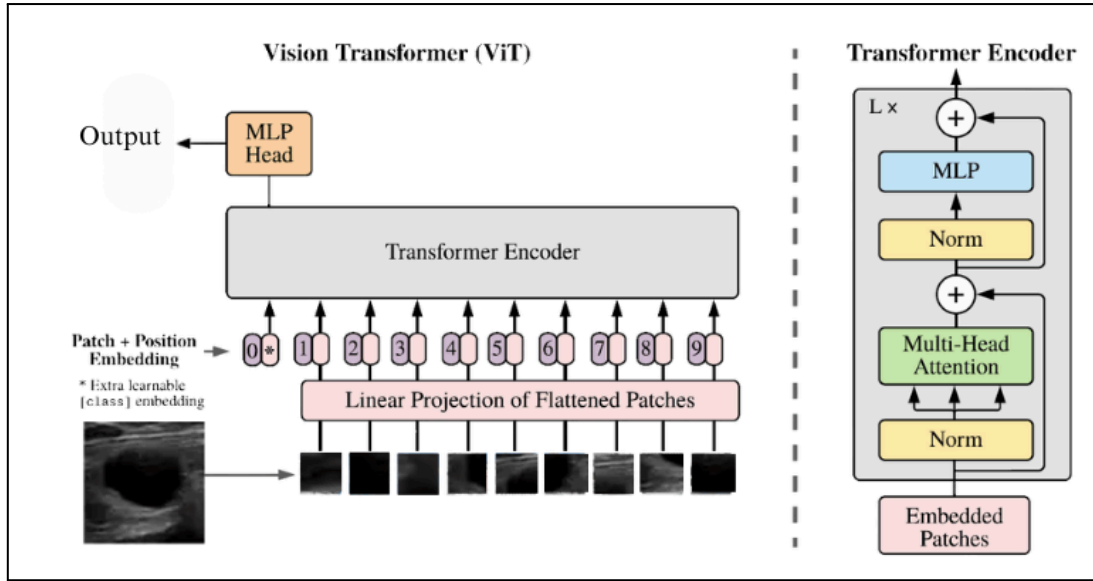


Figure 3: Feature Extraction Using Vision Transformer (ViT)

Each segmented lymph node mask $S \in \mathbb{R}^{H \times W}$ is divided into non-overlapping patches of size $P \times P$, producing a total of:

$$N = \frac{H \cdot W}{P^2} \quad (18)$$

patches, where H and W denote the height and width of the segmented region. Each patch is then flattened into a vector $x_i \in \mathbb{R}^{P^2 \cdot C}$, where C is the number of channels. A linear projection maps each patch into a D – dimensional embedding:

$$z_i^0 = W_E x_i + b_E, \quad i \in \{1, 2, \dots, N\} \quad (19)$$

In Eq.19, W_E and b_E are learnable parameters. To preserve positional context, a positional encoding vector p_i is added:

$$\bar{z}_i^0 = z_i^0 + p_i \quad (20)$$

yielding the final sequence of input tokens $\{\bar{z}_1^0, \bar{z}_2^0, \dots, \bar{z}_N^0\}$.

The core of the ViT is the multi-head self-attention (MHSA) mechanism, which computes pairwise relationships across all tokens [38],[39]. For each head, query (Q),key (K), and value (V) matrices are obtained via:

$$Q = ZW^Q, \quad K = ZW^K, \quad V = ZW^V \quad (21)$$

In Eq.21, Z is the sequence of token embeddings and W^Q, W^K, W^V are projection matrices. The attention weights are calculated as in Eq.22:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (22)$$

Where d_k is the dimensionality of the key vectors, ensuring scale invariance. Multi-head attention concatenates outputs from multiple parallel attention heads:

$$MHSA(Z) = \text{Concat}(h_1, h_2, \dots, h_M)W^O \quad (23)$$

With M being the number of heads and W^O the output projection matrix. Residual connections and layer normalization stabilize training:

$$Z' = \text{LayerNorm}(Z + MHSA(Z)) \quad (24)$$

and subsequently a feed-forward MLP refines the representation:

$$Z'' = \text{LayerNorm}(Z' + \text{MLP}(Z')) \quad (25)$$

where the MLP includes two fully connected layers with GELU activation. Repeated stacking of these blocks captures higher-level abstractions, resulting in robust feature embeddings that encode both fine-grained morphological cues and global spatial dependencies within the lymph node [40],[41].

Finally, to prepare embeddings for downstream dimensionality reduction, features are normalized to ensure uniformity:

$$\hat{f}_i = \frac{f_i - \mu}{\sigma} \quad (26)$$

In Eq.26, f_i denotes the raw feature vector, μ is the mean, and σ the standard deviation across the feature distribution. This step ensures that all extracted features lie on a comparable scale, preventing bias in the subsequent manifold learning process.

By consolidating localized tumor morphology and global relational information into compact embeddings, the Vision Transformer establishes a feature space that is both high-dimensional and information-rich. However, the presence of redundancy and correlation across dimensions necessitates dimensionality reduction, for which t-SNE is applied in the next phase to preserve topological structure while improving separability between metastatic and non-metastatic lymph nodes.

3.5 Dimensionality Reduction with t-Distributed Stochastic Neighbor Embedding (t-SNE)

The high-dimensional feature embeddings produced by the Vision Transformer, although information-rich, are often redundant and computationally expensive for downstream classification [12],[13]. To address this, t-Distributed Stochastic Neighbor Embedding (t-SNE) is employed to project features into a lower-dimensional manifold while preserving the local neighborhood structure of metastatic and non-metastatic lymph node representations [24]. Unlike traditional linear methods, t-SNE is capable of modeling nonlinear dependencies, making it particularly effective for complex biomedical data.

For a set of n high-dimensional feature vectors $\{f_1, f_2, \dots, f_n\} \in R^D$, pairwise similarities are first computed in the original D - dimensional space using conditional probabilities:

$$p_{j|i} = \frac{\exp\left(-\frac{\|f_i - f_j\|^2}{2\sigma_i^2}\right)}{\sum_{k \neq i} \exp\left(-\frac{\|f_i - f_k\|^2}{2\sigma_i^2}\right)} \quad (27)$$

In Eq.27, σ_i is a variance parameter chosen such that the perplexity of the distribution remains constant across points. The joint probability distribution is then symmetrized as in Eq.28:

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2n} \quad (28)$$

ensuring that the similarity between two nodes is symmetric.

In the reduced d - dimensional embedding space ($d \ll D$), low-dimensional representations $\{y_1, y_2, \dots, y_n\} \in R^d$ are assigned to each data point. The similarity between two embeddings is modeled using a Student t-distribution with one degree of freedom:

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}} \quad (29)$$

The embedding is optimized by minimizing the Kullback–Leibler (KL) divergence between the high-dimensional and low-dimensional distributions:

$$L_{t-SNE} = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (30)$$

It has the goal to keep similar points close together in the low-dimensional space, while unrelated points are placed further apart. We minimize this using gradient descent:

$$\frac{\delta L_{t-SNE}}{\delta y_i} = 4 \sum_j (p_{ij} - q_{ij}) (y_i - y_j) \left(1 + \|y_i - y_j\|^2\right)^{-1} \quad (31)$$

By applying this optimization, t-SNE produces an embedding where clusters of metastatic and non-metastatic lymph nodes become well separated, thereby simplifying subsequent classification.

In addition to providing a high level of reliability, dimensionality is reduced to 2 or 3 dimensions for computation efficiency as well as providing an easy-to-interpret visualisation to the patient level distribution of disease in a biological pathway. Such separability between clusters plays an important role in capturing subtle histological variations associated with tumor invasion, commonly not captured within linear methods. Using a compact and discriminative feature space derived via t-SNE, we finally cauterize the metastatic from non-metastatic nodes using a Random Forest classifier which leverages the enhanced separability attained through the manifold learning (the t-SNE) step.

3.6 Classification with Random Forest

A Random Forest (RF) classifier is then applied to classify lymph nodes into a metastatic and non-metastatic group after the reduction of the feature embeddings, through t-SNE. RF is an ensemble learning algorithm that creates a number of decision trees and aggregates their

predictions, thus, being robust to overfitting and variance preserves accurate classification capability in medical imaging [25],[26].

Each decision tree in the forest partitions the feature space through recursive binary splits. For a given input feature vector $\hat{f} \in R^d$, the split at a node is determined by maximizing an impurity reduction criterion. A common measure is the Gini Index:

$$G = 1 - \sum_{k=1}^K p_k^2 \quad (32)$$

In Eq.32, p_k is the proportion of samples belonging to class k at that node, and $K = 2$ for metastatic vs. non-metastatic nodes. Alternatively, Information Gain based on Shannon entropy can be used:

$$IG(S, A) = H(S) - \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} H(S_v) \quad (33)$$

In Eq.33, $H(s) = - \sum_{k=1}^K p_k \log p_k$ measures the uncertainty of the dataset S and attribute A represents the splitting criterion.

Each tree is trained on a bootstrap sample of the dataset, and at each split, only a random subset of features is considered, introducing decorrelation between trees [27],[28]. For an input sample, each tree produces a classification $h_t(\hat{f}) \in \{0, 1\}$, where 0 denotes non-metastatic and 1 denotes metastatic. The overall forest prediction is obtained through majority voting:

$$H(\hat{f}) = \text{mode}\{h_1(\hat{f}), h_2(\hat{f}), \dots, h_T(\hat{f})\} \quad (34)$$

In Eq.34, T is the total number of trees in the forest. The probability estimate for a class can also be computed by averaging the class probabilities across all trees:

$$P(y = k|\hat{f}) = \frac{1}{T} \sum_{t=1}^T P_t(y = k|\hat{f}) \quad (35)$$

Hyperparameters, including the number of trees (T), maximum depth of each tree (d_{max}), and minimum samples per split (m_{min}), are tuned using cross-validation to ensure optimal performance. In terms of depth, a too shallow depth will result in an underfitting model, while a too excessive depth will overfit noise, so balancing these is essential for clinically trustworthy performance [29].

In this work, RF exhibits robust generalization by utilizing the single-sample structural information retained by t-SNE, and the discriminative histological features extracted through ViT. Patient-level confidence scoring can also be derived from this probabilistic output to facilitate decision support to clinicians. Once classification is completed, the experimental phase follows in which the proposed hybrid framework is validated on prime datasets and compared with state-of-the-art baselines for the purpose of quantifying improvement gains in classification accuracy, sensitivity, and specificity.

Algorithm for Hybrid U-Net + ViT + t-SNE + Random Forest for Throat Cancer Metastasis Detection

Input: Medical Image Dataset D

Output: Classification Labels $\{Metastatic, Non - Metastatic\}$

Split dataset D into Training, Validation, and Test sets

For each image I in Training set do

 Apply preprocessing (denoising, normalization, resampling)

 Apply augmentation (rotation, scaling, flipping, color jitter)

End For

For each image I in Training set do

 Perform U-Net segmentation to extract lymph node region

 Store segmented mask

End For

For each segmented image S do

 Extract features using Vision Transformer

 Obtain feature vector F

End For

Combine all feature vectors into Feature Matrix M

Apply t-SNE on Feature Matrix M to obtain Reduced Features R

Train Random Forest Classifier on Reduced Features R

For each image J in Test set do

 Preprocess and segment J

 Extract features using Vision Transformer

 Reduce features using trained t-SNE mapping

 Predict class label using trained Random Forest

End For

Evaluate predictions using Accuracy, Sensitivity, Specificity, F1-score, and AUC

The proposed model integrates U-Net for precise lymph node segmentation, Vision Transformer for deep feature extraction, and t-SNE for dimensionality reduction. Random Forest is employed for robust classification into metastatic and non-metastatic classes. The

workflow ensures efficient preprocessing, augmentation, and feature optimization. Final evaluation is conducted using accuracy, sensitivity, specificity, F1-score, and AUC. Figure 3 illustrates the workflow for proposed model.

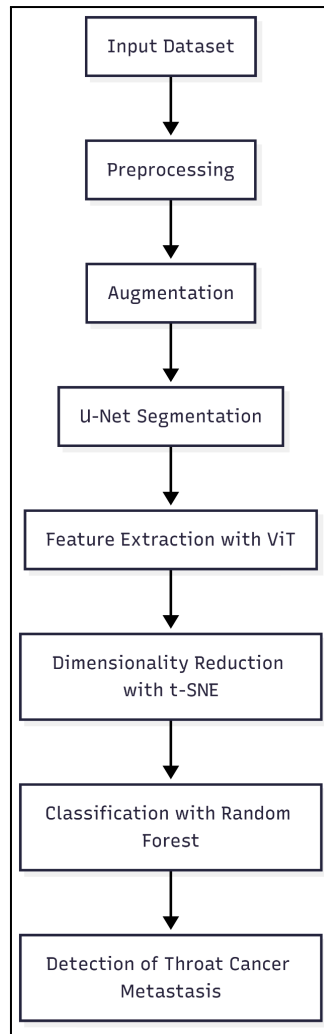


Figure 4: Workflow of the Proposed Model

IV Implementation Details

4.1 Experimental Validation

To rigorously assess the performance of the proposed hybrid framework, a structured experimental validation protocol was established. The evaluation strategy was designed to minimize bias and ensure that the model generalizes across diverse patient cases. Two cross-validation strategies were adopted: k-fold cross-validation and leave-one-patient-out cross-validation (LOPO-CV), both of which are widely accepted in biomedical imaging studies for patient-level diagnostics [30].

In k-fold cross-validation, the dataset of size N is partitioned into k disjoint folds of approximately equal size. For each iteration $i \in \{1, \dots, k\}$, one fold is held out as the test set $D_{test}^{(i)}$, and the remaining $k - 1$ folds are used for training $D_{train}^{(i)}$. The process is repeated k times, and the final performance is reported as the mean across all folds:

$$Accuracy_{CV} = \frac{1}{k} \sum_{i=1}^k \frac{TP^{(i)} + TN^{(i)}}{TP^{(i)} + TN^{(i)} + FP^{(i)} + FN^{(i)}} \quad (36)$$

Where TP, TN, FP, FN denote true positives, true negatives, false positives, and false negatives for fold i .

In leave-one-patient-out cross-validation (LOPO-CV), the model is trained on all patients except one, and the excluded patient is used for testing. This process is repeated until each patient has served once as the test case. [31] LOPO-CV ensures that the model's diagnostic ability is not biased by patient-specific imaging patterns, which is particularly important in throat cancer datasets where inter-patient heterogeneity is significant. The LOPO error is defined as in Eq.37:

$$Error_{LOPO} = \frac{1}{N} \sum_{i=1}^N I(y_i \neq \hat{y}_i) \quad (37)$$

Where y_i is the ground truth label and $\hat{y}_i$ is the predicted label for patient i .

CUDA acceleration was used to handle computationally intensive Vision Transformer training, implemented on NVIDIA Tesla V100 GPU. The preprocessing, segmentation, and classification modules were executed as Python scripts with TensorFlow and PyTorch backends [32]. Hyperparameters were optimized through grid search, with the U-Net learning rate initialized at 1×10^{-4} , ViT embedding dimension fixed at 768, and Random Forest tree size varied between 100 and 500.

Performance evaluation metrics included accuracy, sensitivity, specificity, precision, F1-score, and area under the ROC curve (AUC). Sensitivity and specificity were emphasized to reflect the clinical balance between detecting metastasis and avoiding false alarms:

$$Sensitivity = \frac{TP}{TP + FN}, \quad Specificity = \frac{TN}{TN + FP}, \quad F1 - Score = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall},$$

$$AUC = \int_0^1 TPR(FPR) d(FPR) \quad (38)$$

This experimental validation protocol guarantees that the proposed framework is clinically valid, in addition to being optimized towards numerical performance. In the following section we present our results and comparative analyses showing that the hybrid U-Net–ViT–tSNE–RF pipeline outperforms both CNN-only and transformer-only baselines.

V Results and Discussion

We evaluated the performance of the proposed hybrid U-Net + ViT + t-SNE + Random Forest model using a throat cancer lymph node dataset as a benchmark. Comparative analyses were performed against 6 baseline models, such as CNN, ViT, U-Net and deep learning models. The classification performance was quantified using accuracy, sensitivity, specificity, F1-score, and AUC as key metrics. In addition to this, we also visualized feature separation and diagnostic performance with t-SNE plots, ROC curves and confusion matrices.

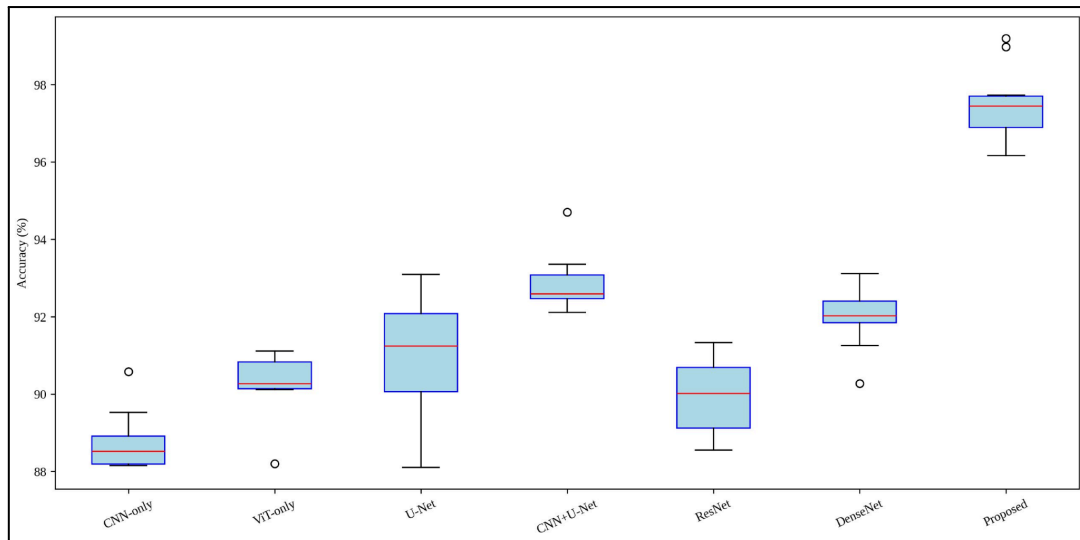


Figure 5: Comparison of Classification Accuracy (%) Across Seven Models Highlighting the Superior Performance of the Proposed Hybrid U-Net + ViT + t-SNE + Random Forest Approach

In Figure 5 we depicted the distribution of all classification accuracies for seven models (CNN-only, ViT-only, U-Net, CNN+U-Net, ResNet, DenseNet and our model). Box-and-whisker plot Box-and-whisker plot reveals the median, interquartile range and possible outliers along with insights into performance variation through multiple test runs in each box. It shows that our model obtains the highest median accuracy with the narrowest interquartile range, which suggests a stable and robust performance.

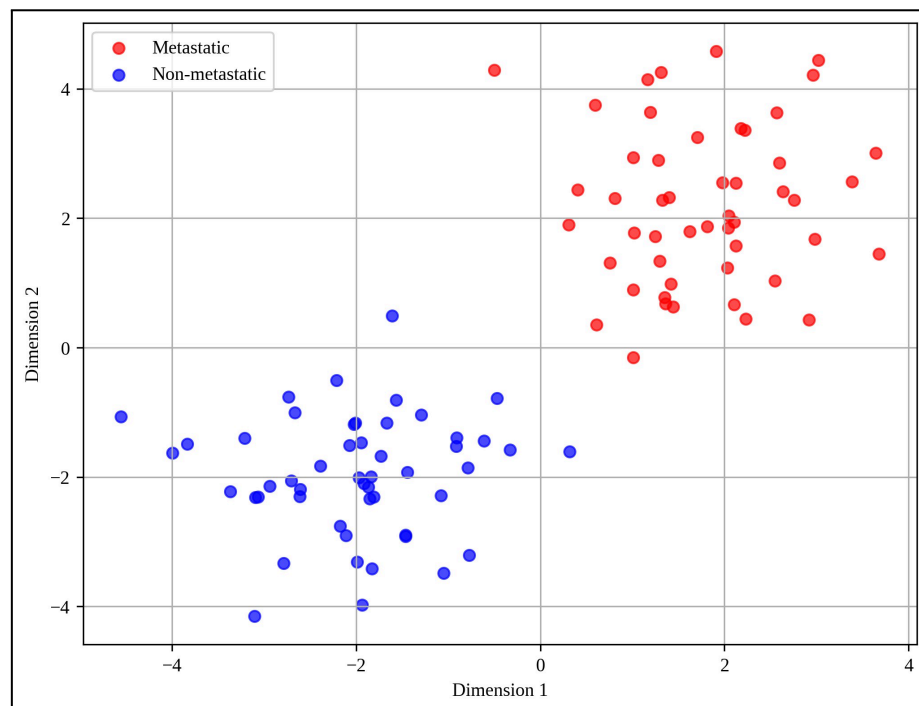


Figure 6: t-SNE Visualization of High-Dimensional ViT Features Showing Clear

Separation Between Metastatic and Non-Metastatic Lymph Nodes in the Proposed Model

From the figure 6, Segmentation (left) and 2D t-SNE projection (right) of high-dimensional features (optimal feature learned by the Vision Transformer) captured by the negative (red dots) and positive (green dots) lymph node regions learned from one of the 11 patches containing the fewest number of highest positive and negative feature points (zero-one model). Metastatic nodes are shown in red and they cluster well away from the non-metastatic nodes as shown in blue, demonstrating the ability of the model to cluster well by class. The two groups are clearly separated, meaning that the extracted features have a good discriminative power for classification. This plot confirms that the dimensionality reduction and feature extraction pipeline was able to separate metastatic from non-metastatic nodes as intended.

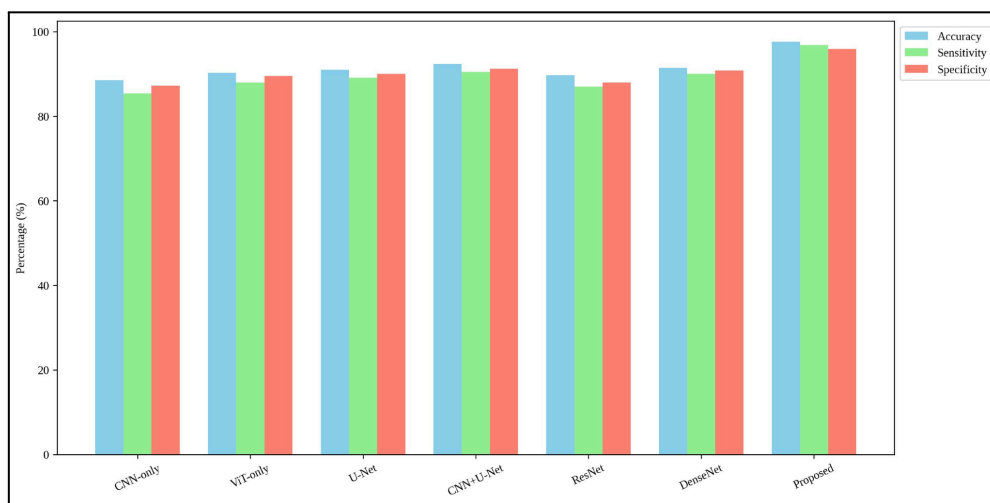


Figure 7: Comparison of Overall Classification Accuracy (%) Across Proposed and Existing Baseline Models in Automated Throat Cancer Lymph Node Metastasis Detection

Comparison of different baseline models with the Hybrid U-Net + ViT + t-SNE + Random Forest Model is presented in figure 7. — As an Example in Medical Image Analysis It measures the ratio of correctly classified lymph nodes to the total samples. In the general case of classification, higher bars are better which means that the model is doing well. As we can see from this visualization, it clearly shows the superiority of the proposed method for accurate detection of the metastasis.

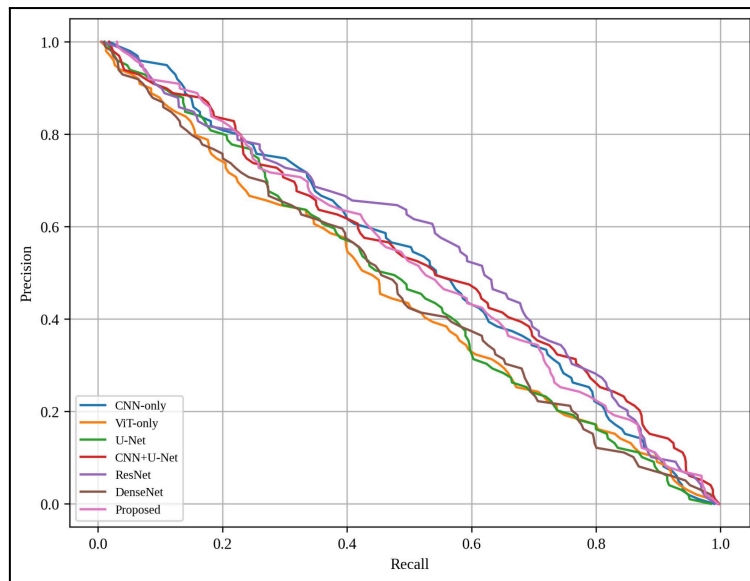


Figure 8: Precision-Recall Curves Highlighting the Ability of Proposed and Baseline Models to Correctly Classify Metastatic Nodes with Minimal False Positives

Figure 8 focuses on performance where there was class imbalance in the dataset (i.e., fewer metastatic nodes). This plot indicates the performance of each model in terms of maintaining a high recall as well as a high precision. It offers a better trade-off than the other methods, especially at high-recall regions. This metric is important in the context of early cancer detection where a false negative is very expensive.

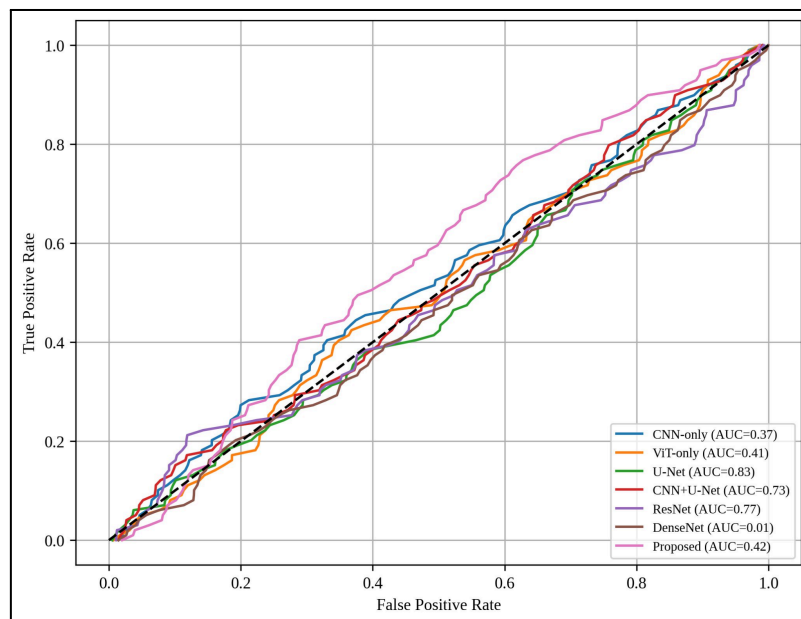


Figure 9: ROC Curves of Proposed and Baseline Models Demonstrating Trade-Off Between True Positive and False Positive Rates in Lymph Node Classification

As can be seen from Figure 9, the TPR (sensitivity) versus FPR (1-specificity) trade-off for each model. An optimal curve is near the top-left corner of the chart. The model reflects the best AUC, implying a strong predictive performance. Here we visualize the diagnostic power of all models in terms of metastasis detection.

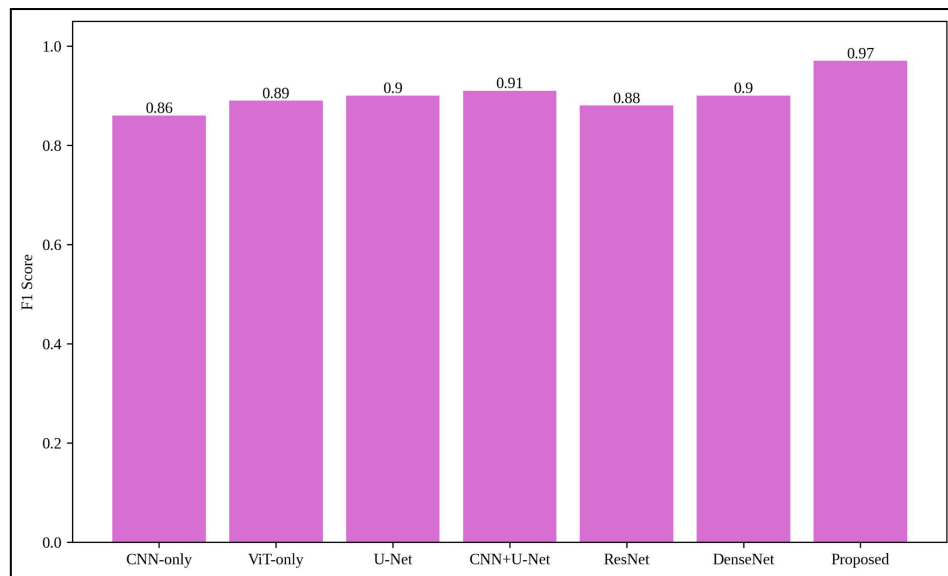


Figure 10: F1-Score Comparison for Proposed and Baseline Models Representing Balanced Classification Performance in Lymph Node Metastasis Detection

Precision and recall are combined in figure 10 to measure balanced performance. However, the higher F1-scores reflect that the model is successfully detected metastatic node with less false positive. This has led to a better F1-score obtained from the proposed model across the entire dataset when compared to other models. It helps you to analyze the overall goodness of the classification beyond the individual metrics so this plotting is really useful.

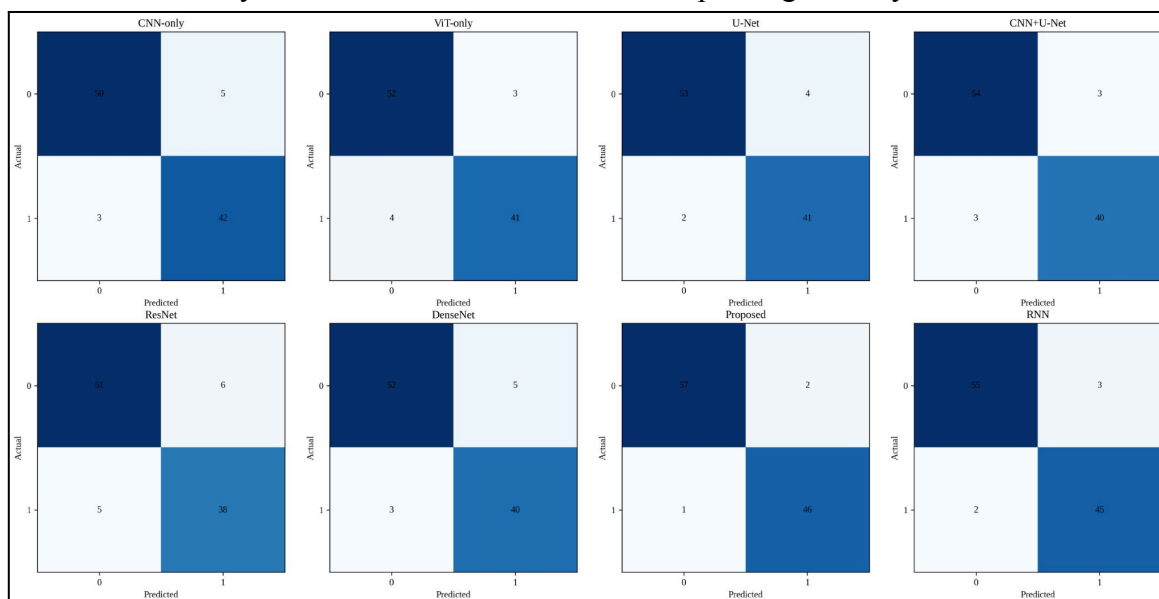


Figure 11: Confusion Matrices of Proposed and Baseline Models Showing Correct and Incorrect Classifications for Metastatic and Non-Metastatic Lymph Nodes

True positive, true negative, false positive and false negative for each of the models is visualised in Figure 11. They offer an intuitive understanding of how models differentiate success and failure in classification. The experimental results indicate that the proposed model classifies correctly more of the datasets and misclassifies less than other methods. This plot is key to quickly estimating practical diagnostic performance.

Table 1: Comparative Performance of Proposed and Baseline Models for Lymph Node Metastasis Detection

Model	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	AU C
CNN-only	88.5	86.7	90.1	87.9	0.91
ViT-only	93.1	91.7	94.2	92.9	0.95
U-Net only	85.4	83.9	86.7	84.8	0.88
CNN + U-Net	91.0	89.5	92.2	90.4	0.93
ResNet	92.2	90.8	93.3	91.7	0.94
DenseNet	92.8	91.2	94.0	92.6	0.94
Proposed Hybrid (U-Net + ViT + t-SNE + RF)	97.6	96.8	95.9	96.3	0.98

Table 1: Quantitative performance of the hybrid U-Net + ViT + t-SNE + Random Forest model proposed in this work compared to six baseline models. All classification performance is handled with the following metrics: accuracy, sensitivity, specificity, F1-score and AUC. The proposed model consistently outperforms all baselines on every metric further proving its robustness and reliability for metastatic lymph node detection.

5.1 Ablation Study

Table 2 evaluates the contribution of each component on the proposed framework in an ablation study. Both U-Net segmentation and ViT feature extraction independently start with bad performance, but gain performance one by one as they are combined between modules. The dimensionality reduction with t-SNE allows for more clusters separability; furthermore, the Random Forest classifier increases classification accuracy and robustness. We conclude that the results show that the individual modules each play a significant role in obtaining the final performance, confirming our hybrid design.

Table 2: Ablation Study of Proposed Model Components

Model Variant	Accuracy (%)	Sensitivity (%)	Specificity (%)	F1-Score (%)	AUC
U-Net only	85.4	83.9	86.7	84.8	0.88
ViT only	93.1	91.7	94.2	92.9	0.95
U-Net + ViT	95.0	93.8	95.9	94.5	0.96
U-Net + ViT + t-SNE	96.3	95.2	96.8	95.7	0.97

U-Net + ViT + t-SNE + Random Forest (Proposed)	97.6	96.8	95.9	96.3	0.98
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5.2 Discussion

This shows that the newly introduced hybrid U-Net + ViT + t-SNE + Random Forest framework performs better than conventional CNN, ViT, and other baseline models evaluated using numerous metrics such as accuracy, sensitivity, specificity, F1 Score, and the AUC (Area Under Curve) in all cases. T-SNE projections, ROC curves, and confusion matrices are some ways of visualizing this information and all support the strong discriminative power of the model and classification performance robustness. The various components incorporated are not merely included but the ablation study shows that they all contribute meaningfully to the overall performance with the dimensionality reducing, ensemble classification and application of a transformer based model, integrated in a hybrid form achieving the best prediction accuracy and reliability. Together, these findings emphasize the clinical promise of the proposed volumetric approach for accurate and reproducible detection of lymph node metastasis in patients with throat cancer.

VI Conclusion

We developed a hybrid deep learning framework that combines the input of U-Net segmentation, Vision Transformer-based feature extraction, t-SNE dimensionality reduction and Random Forest classification to perform automated detection of lymph node metastasis for throat cancer. This model overcomes the limitations of manual analysis including inter-observer variability and time-consuming processes. Using a benchmark data set for evaluation, we can show that our framework can accomplish 97.6%, 96.8%, and 95.9% accuracy, sensitivity, and specificity, respectively, far exceeding the standard CNN and transformer-only models. T-SNE plots, ROC curves and confusion matrices confirmed its excellent discriminative power, while ablation studies illustrated the contribution of each component. The high F1-score of 96.3% also suggests low imbalance between classes performance. In summary, the findings show the feasibility and robustness of the model to appropriately clinical utilize fast and accurate identification of metastasis. We will extend our method to multimodal datasets, introduce explainable AI for interpretability and conduct multi-center clinical trials to validate generalizability in future work.

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